THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 6 cm. The length of each tangent is 8 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 28 cm **(b)** 16 cm **(c)** 32 cm **(d)** 14 cm

2. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 100^\circ$. The $\angle APB$ is: 1

(a) 100° **(b)** 40° **(c)** 200° **(d)** 80°

3. PA is a tangent at A to a circle with centre O, and $\angle OPA = 35^\circ$. The $\angle AOP$ is: 1

(a) 55° **(b)** 125° **(c)** 35° **(d)** 145°

4. The angle between a tangent to a circle and the radius through the point of contact is: 1

(a) 45° **(b)** 90° **(c)** 180° **(d)** 0°

5. A point P is 13 cm from the centre O of a circle of radius 5 cm. The length of the tangent drawn from P to the circle is: 1

(a) 144 cm **(b)** 8 cm **(c)** 12 cm **(d)** 18 cm

6. The length of the tangent from an external point P to a circle is 12 cm and P is 15 cm from the centre. The radius of the circle is: 1

(a) 9 cm **(b)** 3 cm **(c)** 27 cm **(d)** 13 cm

7. A tangent to a circle of radius 10 cm has length 24 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 14 cm **(b)** 34 cm **(c)** 26 cm **(d)** 52 cm

8. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 5 cm. The length of each tangent is 12 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 24 cm **(b)** 17 cm **(c)** 34 cm **(d)** 32 cm

9. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 130^\circ$. The $\angle APB$ is: 1

(a) 25° **(b)** 260° **(c)** 130° **(d)** 50°

10. PA is a tangent at A to a circle with centre O, and $\angle OPA = 40^\circ$. The $\angle AOP$ is: 1

(a) 130° **(b)** 40° **(c)** 50° **(d)** 140°

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A tangent to a circle of radius 8 cm has length 6 cm from the external point P to the point of contact. Find the distance of P from the centre of the circle. 2

12. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 90^\circ$. Find the $\angle APB$. 2

13. A point P is 29 cm from the centre O of a circle of radius 21 cm. Find the length of the tangent drawn from P to the circle. 2

14. A tangent to a circle of radius 12 cm has length 5 cm from the external point P to the point of contact. Find the distance of P from the centre of the circle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 15 cm. The length of each tangent is 20 cm. Find the perimeter of the quadrilateral $PAOB$. 3

16. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 5 cm. The length of each tangent is 12 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

17. PA is a tangent at A to a circle with centre O, and $\angle OPA = 50^\circ$. Find the $\angle AOP$. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 10 cm. The length of each tangent is 24 cm. Find the perimeter of the quadrilateral $PAOB$. 3

18. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 9 cm. The length of each tangent is 12 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

OR

PA is a tangent at A to a circle with centre O, and $\angle OPA = 60^\circ$. Find the $\angle AOP$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Tangents PA and PB are drawn from a point P at a distance of 13 cm from the centre O of a circle of radius 5 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 26 cm and 10 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

20. Tangents PA and PB are drawn from a point P at a distance of 15 cm from the centre O of a circle of radius 9 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 29 cm and 20 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

END OF QUESTION PAPER — **Total: 40 Marks**